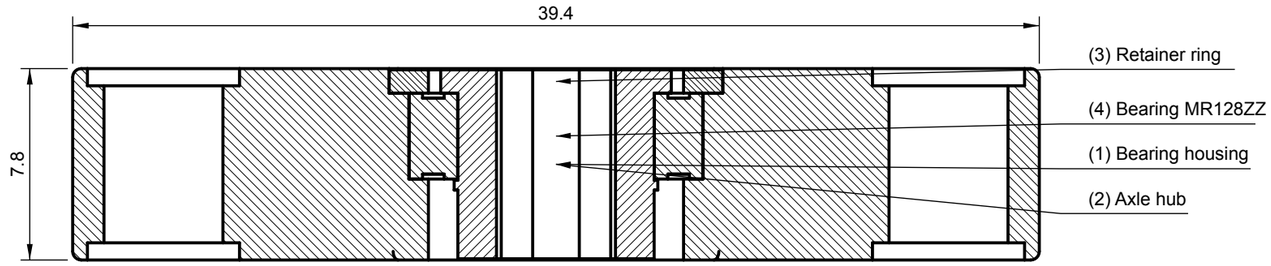
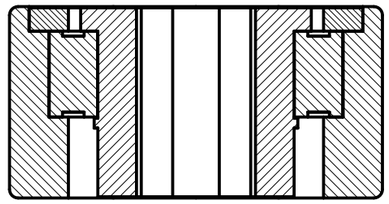
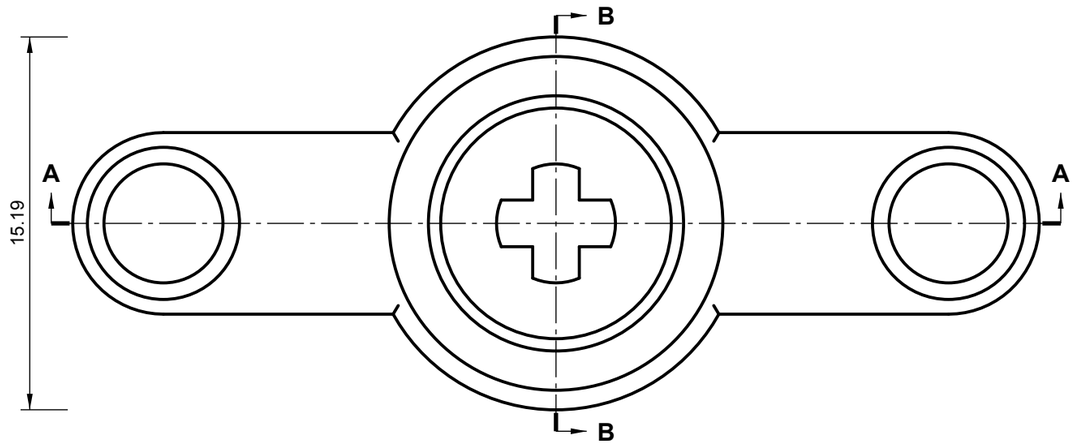
A–A



B–B

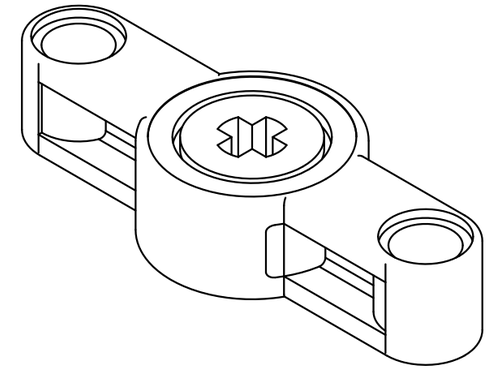




Bill of materials 明细表

#	Part No. 零件号	Name 名称	Material 材料	Colour	Qty	g
1	SB-BR5-01	Bearing housing 轴承座	PA66-GF30	Teal	1	1.76
2	SB-BR5-02	Axle hub 十字轴芯	POM copolymer	Light grey	1	0.42
3	SB-BR5-03	Retainer ring 压圈	POM copolymer	Teal	1	0.08
4	purchased 外购	Bearing MR128ZZ 轴承	Chrome steel GCr15	Bright steel	1	1.64

ISOMETRIC 轴测图 (2.5:1, ref.)



Technical requirements 技术要求

1. Assembly drawing: components per BOM, each with its own drawing (sheets 2…).
装配图：零件见明细表，各零件另有图纸（第 2 页起）。

2. Press the bearing into the housing by its outer ring only, then press in the retainer ring. Press the hub into the bearing by its inner ring only until the snap bead clicks under it. Never push through the balls.
轴承仅压外圈装入轴承座，再压入压圈；轴芯仅压内圈装入轴承，直至卡圈卡住。不得经滚珠传力。

3. 100% spin check: hub turns freely by finger, no grinding. Sample test (proposed): hub push-out ≥ 50 N, bearing push-out ≥ 80 N.
100% 转动检查：手拨轴芯顺畅、无异响。抽检（建议值）：轴芯推出力≥50 N，轴承推出力≥80 N。

4. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

5. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

6. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	PA66GF + POM + STEEL see BOM 见明细表	**Weight** 重量	3.90 g
Colour 颜色	see BOM 见明细表	**Qty / kit** 每套数量	4
Finish 表面	see BOM 见明细表	**Scale** 比例	4:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	1 / 4

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
A	First issue for quotation & samples 首次发布 · 报价及打样	2026-10-27

Drawn 设计	semiblock.ai	2026-10-27
Checked 审核		
Approved 批准		

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Part No. 零件号

SB-BR5

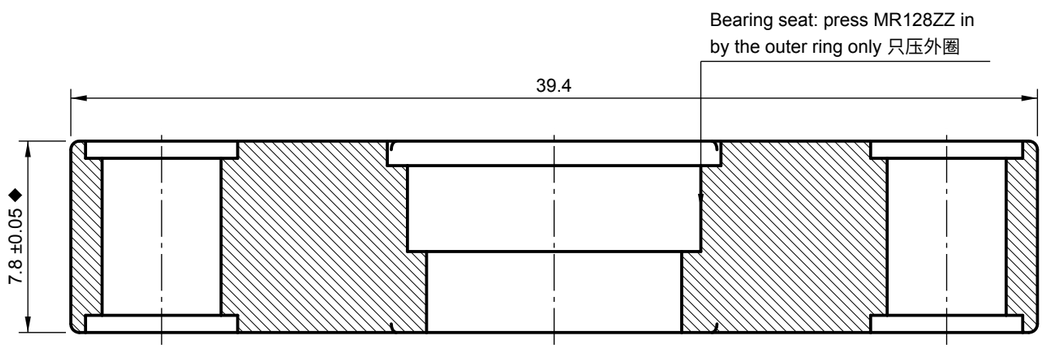
Name 名称

Bearing beam 1x5 (ball bearing) — assembly 装配图

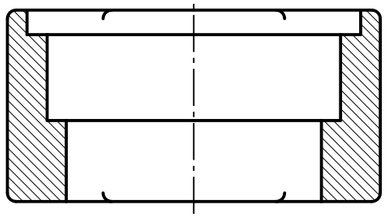
轴承梁 1×5（滚珠轴承）

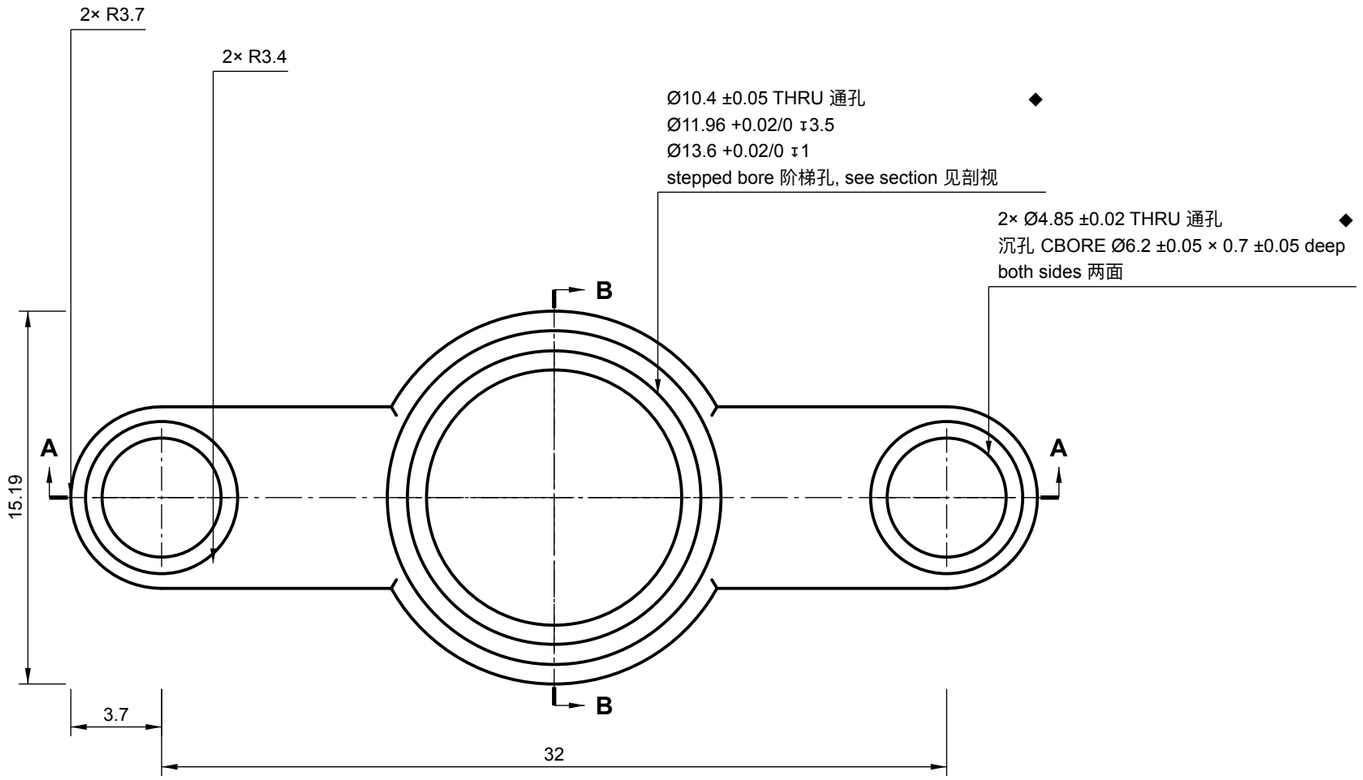
Dwg SB-BR5 · Rev A · A3 · 3D: SB-BR5.step / .FCStd

A–A

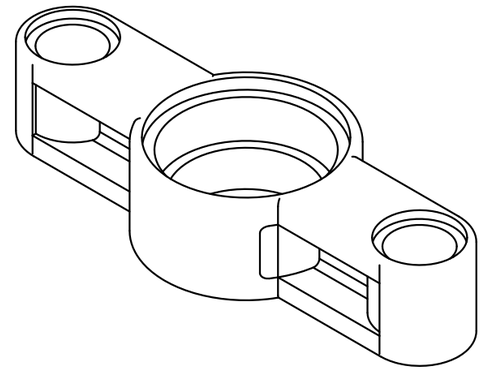


B–B





ISOMETRIC 轴测图 (2.5:1, ref.)



Technical requirements 技术要求

1. 3D model SB-BR5.step is the master; dimensions not shown are taken from it.
以 3D 模型 SB-BR5.step 为准，未注尺寸按 3D 数据。

2. Material PA66-GF30, e.g. Zytel 70G30 or equiv.; colour Teal 青绿 (sRGB #12A89A, std SB-C08); colour plaque approved before production.
材料：PA66+30%玻纤（杜邦 70G30 或同等）；颜色 青绿（#12A89A，色号 SB-C08），量产前确认色板。

3. Draft ≤ 0.5° on outer walls, added by the tool maker within tolerance. Pin / axle holes and ◆ features: draft ≤ 0.25°, ◆ size held over the full depth.
外壁拔模≤0.5°（模具厂在公差内添加）；销孔、十字孔及◆特征拔模≤0.25°，全深度保持◆尺寸。

4. Parting line, gate and ejector marks: not on visible faces or inside holes, flush to +0.05. Positions to be agreed in DFM.
分模线、浇口、顶针痕不得在外观面及孔内，凸起≤0.05，位置于 DFM 时确认。

5. No flash > 0.05, sink marks, short shots, silver streaks or voids. Holes free of burrs.
无大于0.05的披锋，无缩水、缺料、银纹、气泡；孔内无毛刺。

6. Unspecified edge radii R0.2–R0.35 as in 3D. No sharp edges or points (toy safety).
未注圆角 R0.2–R0.35（按 3D），不得有锐边锐角（玩具安全）。

7. Pin holes: GO Ø4.83 / NO-GO Ø4.88 plug gauge. Fit test with SemiBlock and LEGO® Technic™ pins; push-in / pull-out force agreed at T1.
销孔用 Ø4.83 通规 / Ø4.88 止规检验；与 SemiBlock 及乐高销做装配测试，插拔力于 T1 确认。

8. Mark cavity number and material code >PA66-GF30< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >PA66-GF30<（ISO 11469），空间不足可省略。

9. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

10. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

11. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	PA66-GF30 PA66+30%玻纤	**Weight** 重量	1.76 g
Colour 颜色	Teal 青绿 #12A89A	**Qty / kit** 每套数量	4
Finish 表面	VDI 3400 Ref 21 细哑光纹理	**Scale** 比例	4:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	2 / 4

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
A	First issue for quotation & samples 首次发布 · 报价及打样	2026-10-27

Drawn 设计	semiblock.ai	2026-10-27
Checked 审核		
Approved 批准		

semiblock.ai SemiBlock Junior

Part No. 零件号

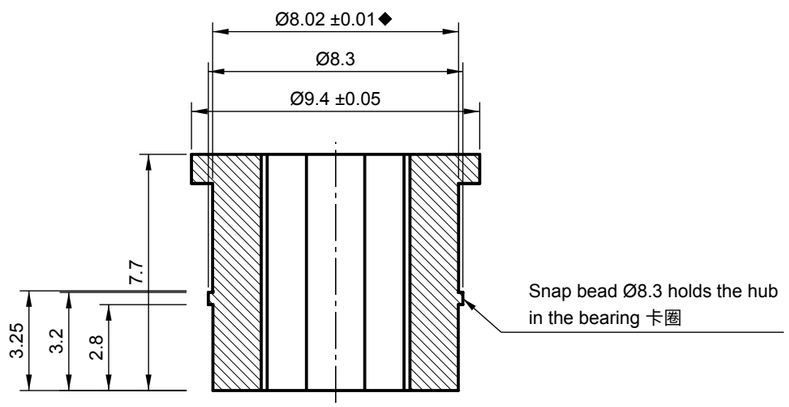
SB-BR5-01

Name 名称

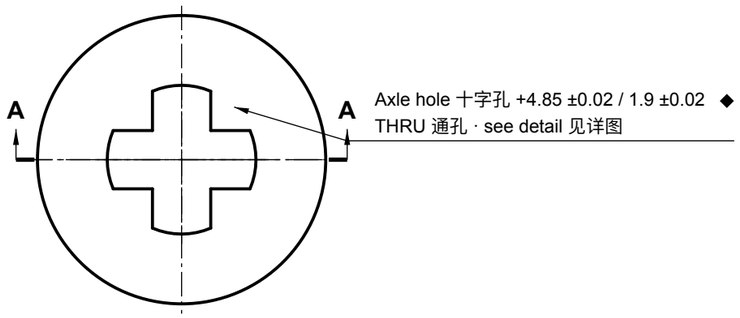
Bearing beam 1x5 (ball bearing) — Bearing housing
轴承梁 1×5（滚珠轴承） — 轴承座

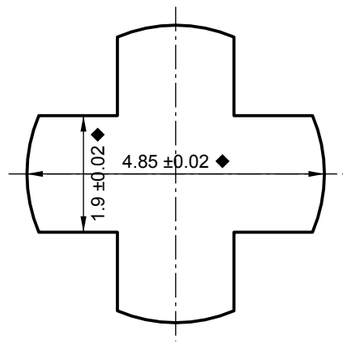
Dwg SB-BR5-01 · Rev A · A3 · 3D: SB-BR5.step / .FCStd

A–A

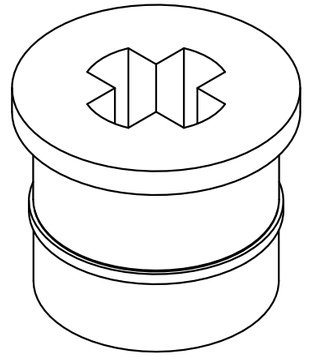


Axle hole 十字孔 (10:1)





ISOMETRIC 轴测图 (5:1, ref.)



Technical requirements 技术要求

1. 3D model SB-BR5.step is the master; dimensions not shown are taken from it.
以 3D 模型 SB-BR5.step 为准，未注尺寸按 3D 数据。

2. Material POM copolymer, e.g. Hostaform C 9021; colour Light grey 浅灰 (sRGB #B4BCC6, std SB-C04); colour plaque approved before production.
材料：POM 共聚甲醛（C 9021 或同等）；颜色 浅灰（#B4BCC6，色号 SB-C04），量产前确认色板。

3. Draft ≤ 0.5° on outer walls, added by the tool maker within tolerance. Pin / axle holes and ◆ features: draft ≤ 0.25°, ◆ size held over the full depth.
外壁拔模≤0.5°（模具厂在公差内添加）；销孔、十字孔及◆特征拔模≤0.25°，全深度保持◆尺寸。

4. Parting line, gate and ejector marks: not on visible faces or inside holes, flush to +0.05. Positions to be agreed in DFM.
分模线、浇口、顶针痕不得在外观面及孔内，凸起≤0.05，位置于 DFM 时确认。

5. No flash > 0.05, sink marks, short shots, silver streaks or voids. Holes free of burrs.
无大于0.05的披锋，无缩水、缺料、银纹、气泡；孔内无毛刺。

6. Unspecified edge radii R0.2–R0.35 as in 3D. No sharp edges or points (toy safety).
未注圆角 R0.2–R0.35（按 3D），不得有锐边锐角（玩具安全）。

7. Axle (+) holes: GO gauge = cross 4.83 / 1.88. Standard axle must slide in by hand without play > 0.05.
十字孔用 4.83/1.88 十字通规；标准十字轴可手推插入，间隙≤0.05。

8. Mark cavity number and material code >POM< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >POM<（ISO 11469），空间不足可省略。

9. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

10. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

11. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	POM copolymer POM 共聚甲醛	**Weight** 重量	0.42 g
Colour 颜色	Light grey 浅灰 #B4BCC6	**Qty / kit** 每套数量	4
Finish 表面	SPI B-2 光滑面	**Scale** 比例	5:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	3 / 4

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
A	First issue for quotation & samples 首次发布 · 报价及打样	2026-10-27

Drawn 设计	semiblock.ai	2026-10-27
Checked 审核		
Approved 批准		

semiblock.ai — SemiBlock Junior

Part No. 零件号

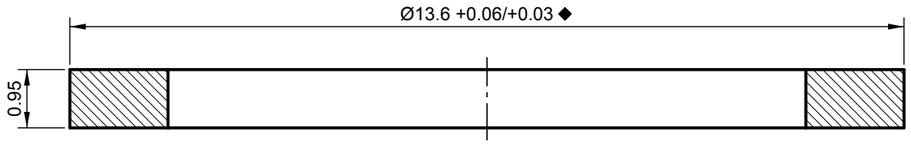
SB-BR5-02

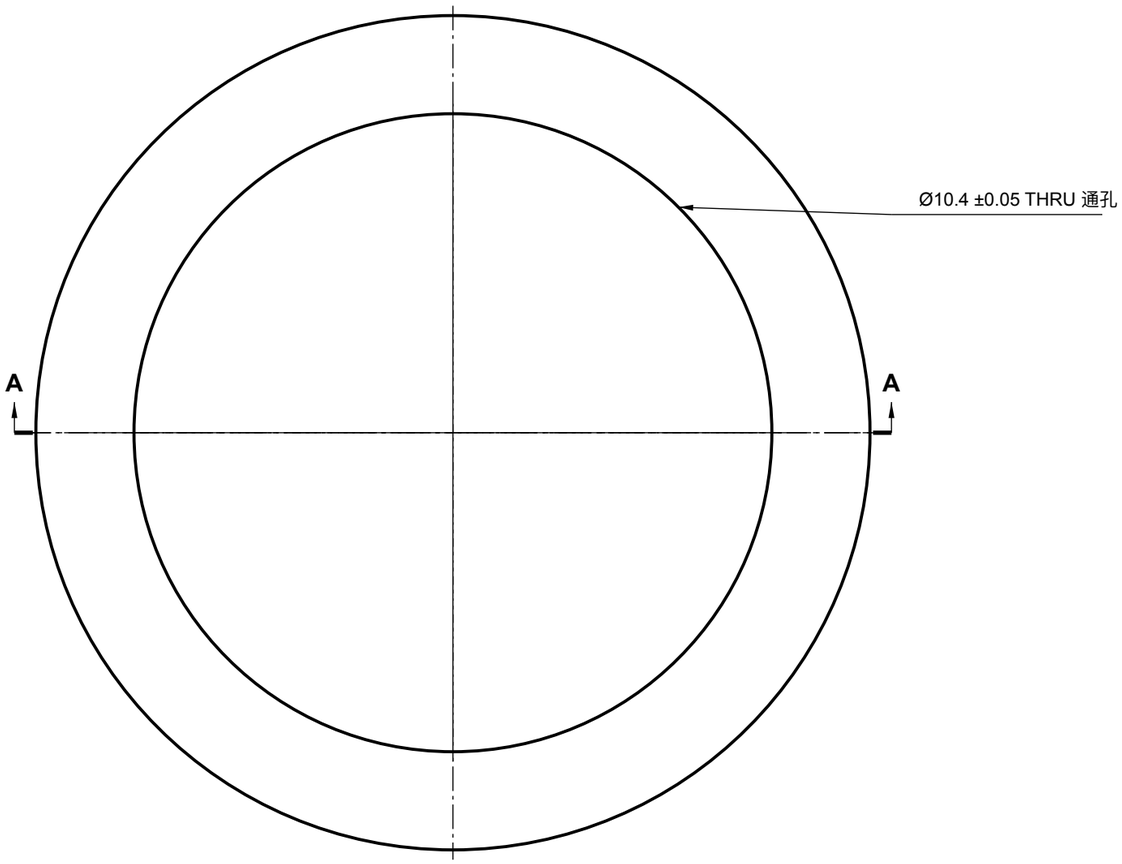
Name 名称

Bearing beam 1x5 (ball bearing) — Axle hub
轴承梁 1×5（滚珠轴承） — 十字轴芯

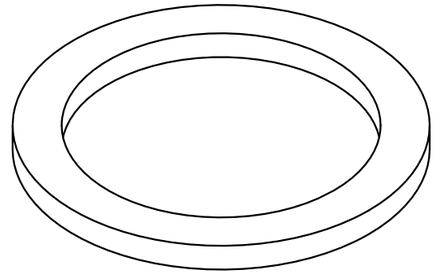
Dwg SB-BR5-02 · Rev A · A3 · 3D: SB-BR5.step / .FCStd

A–A





ISOMETRIC 轴测图 (5:1, ref.)



Technical requirements 技术要求

1. 3D model SB-BR5.step is the master; dimensions not shown are taken from it.
以 3D 模型 SB-BR5.step 为准，未注尺寸按 3D 数据。

2. Material POM copolymer, e.g. Hostaform C 9021; colour Teal 青绿 (sRGB #12A89A, std SB-C08); colour plaque approved before production.
材料：POM 共聚甲醛（C 9021 或同等）；颜色 青绿（#12A89A，色号 SB-C08），量产前确认色板。

3. Draft ≤ 0.5° on outer walls, added by the tool maker within tolerance. Pin / axle holes and ◆ features: draft ≤ 0.25°, ◆ size held over the full depth.
外壁拔模≤0.5°（模具厂在公差内添加）；销孔、十字孔及◆特征拔模≤0.25°，全深度保持◆尺寸。

4. Parting line, gate and ejector marks: not on visible faces or inside holes, flush to +0.05. Positions to be agreed in DFM.
分模线、浇口、顶针痕不得在外观面及孔内，凸起≤0.05，位置于 DFM 时确认。

5. No flash > 0.05, sink marks, short shots, silver streaks or voids. Holes free of burrs.
无大于0.05的披锋，无缩水、缺料、银纹、气泡；孔内无毛刺。

6. Unspecified edge radii R0.2–R0.35 as in 3D. No sharp edges or points (toy safety).
未注圆角 R0.2–R0.35（按 3D），不得有锐边锐角（玩具安全）。

7. Mark cavity number and material code >POM< on a hidden face where space allows (ISO 11469).
在不可见面标注穴号及材料代码 >POM<（ISO 11469），空间不足可省略。

8. Toy safety: EN 71-1/-2/-3, ASTM F963, GB 6675.1–.4; RoHS and REACH. Supply material certificates.
玩具安全：EN 71-1/-2/-3、ASTM F963、GB 6675.1–.4；符合 RoHS、REACH，提供材料证明。

9. ◆ = critical dimension: FAI report on every ◆ at T1 (5 pcs per cavity); production Cpk ≥ 1.33.
◆为关键尺寸：T1 每穴 5 件全尺寸首件报告；量产 Cpk≥1.33。

10. Stage-2 drawings are for quotation, DFM review and T0/T1 samples. No mass production before written sample approval.
第二阶段图纸用于报价、DFM 评审及 T0/T1 打样；样品书面确认前不得量产。

Material 材料	POM copolymer POM 共聚甲醛	**Weight** 重量	0.08 g
Colour 颜色	Teal 青绿 #12A89A	**Qty / kit** 每套数量	4
Finish 表面	SPI B-2 光滑面	**Scale** 比例	10:1
Unit 单位	mm	**Projection** 投影	
Status 状态	Rev A quotation 报价及打样	**Sheet** 页	4 / 4

General tolerances 未注公差 (mm)
rigid ABS/POM 硬胶 | TPE 软胶

0 – 6	±0.05	±0.15
6 – 18	±0.08	±0.2
18 – 50	±0.12	±0.3
50 – 120	±0.2	±0.5
angles 角度	±0.5°	±1°

◆ = critical dimension 关键尺寸
FAI 100%, Cpk ≥ 1.33

Rev 版本	Description 修改内容	Date 日期
A	First issue for quotation & samples 首次发布 · 报价及打样	2026-10-27

Drawn 设计	semiblock.ai	2026-10-27
Checked 审核		
Approved 批准		

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Part No. 零件号

SB-BR5-03

Name 名称

Bearing beam 1x5 (ball bearing) — Retainer ring
轴承梁 1×5（滚珠轴承） — 压圈

Dwg SB-BR5-03 · Rev A · A3 · 3D: SB-BR5.step / .FCStd